

ADVANCED ROUTING & INFRASTRUCTURE STUDY PLAN

MODULE 1: NETWORK FOUNDATIONS & ARCHITECTURE

Focus: Control Plane vs Data Plane, RIB/FIB, and CEF.

Explanation:

Modern routers separate the 'Brain' (Control Plane) from the 'Forwarding' (Data Plane). CEF (Cisco Express Forwarding) optimizes performance by pre-calculating paths.

Core Commands:

```
# show ip cef
# show ip route summary
# show processes cpu sorted
```

MODULE 2: ADVANCED OSPF (MULTI-AREA & LSA TYPES)

Focus: LSA Types (1-7), Stubby Areas, and NSSA.

Configuration:

```
router ospf 1
 area 10 nssa no-summary
 area 10 authentication message-digest
 interface Gi0/1
 ip ospf message-digest-key 1 md5 CISC0123
```

MODULE 3: ADVANCED BGP - THE INTERNET PROTOCOL

Focus: Path Selection (Weight, LP, AS-Path, MED) & Communities.

Configuration:

```
router bgp 65001
 neighbor 10.1.1.2 remote-as 65100
 address-family ipv4
   neighbor 10.1.1.2 route-map SET_LOCAL_PREF in
 !
route-map SET_LOCAL_PREF permit 10
 set local-preference 200
```

MODULE 4: VRF-LITE & MULTI-TENANCY

Focus: Route Isolation and Virtual Routing Instances.

Configuration:

```
vrf definition CUSTOMER_A
 rd 100:1
 address-family ipv4
 !
interface Gi0/1.10
 vrf forwarding CUSTOMER_A
```

ADVANCED ROUTING & INFRASTRUCTURE STUDY PLAN

```
ip address 10.1.1.1 255.255.255.0
```

MODULE 5: ROUTE REDISTRIBUTION & MANIPULATION

Focus: Seed Metrics, Route-Tagging, and Filtering.

Explanation:

Always use route-maps during redistribution to prevent routing loops.

Configuration:

```
router ospf 1
 redistribute bgp 65001 subnets route-map BGP_TO_OSPF
 !
route-map BGP_TO_OSPF permit 10
 set tag 555
```

MODULE 6: POLICY-BASED ROUTING (PBR) & IP SLA

Focus: Conditional Forwarding and Performance Tracking.

Configuration:

```
ip sla 1
 icmp-echo 8.8.8.8
 frequency 5
track 1 ip sla 1 reachability
 !
route-map PBR_TRAFFIC permit 10
 match ip address 101
 set ip next-hop verify-availability 10.1.1.2 1 track 1
```

MODULE 7: SECURITY HARDENING (COPP & STIGS)

Focus: Control Plane Policing & Management Security.

Configuration:

```
control-plane
 service-policy input POLICY_COPP_HARDENING
 !
line vty 0 15
 transport input ssh
 ip access-class ACL_VTY_ACCESS in
```

MODULE 8: TELEMETRY & MODERN MONITORING

Focus: Model-Driven Telemetry (gRPC Dial-out).

Explanation:

Real-time streaming vs traditional SNMP polling.

ADVANCED ROUTING & INFRASTRUCTURE STUDY PLAN

show telemetry model-driven subscription 501

MODULE 9: SD-WAN & CLOUD CONNECTIVITY

Focus: Overlay vs Underlay, IPsec Tunnels, and BGP Peering.

Principles:

- Use BGP for Cloud Peering (ExpressRoute/DirectConnect).
- Implement Zscaler for Cloud-Based Security.

MODULE 10: AUTOMATION & PYTHON SCRIPTING

Focus: Netmiko, Ansible, and API-Driven Configs.

Goal: Automated device provisioning and mass configuration.